The Differential: what the simulation showed

West bay 1: the Differential · 4 November 2026

The teeth inside out, with newer OpenSCAD (2026-11-04)

William saw the bevel gears drawn with dark triangular slivers that shimmered as they turned, but only with bpp's shadows off. His OpenSCAD (2026.10.15) made the spider gear inside out: volume -18.5 where it should be +19.7, 11 284 triangles against 932, and not watertight. GregFrost's bevel teeth are each built of small polyhedra whose faces are listed counterclockwise seen from outside; OpenSCAD wants them clockwise. OpenSCAD 2021's CGAL put that right as it joined the pieces; the newer Manifold takes them as given. bpp lights faces by their normals, so these faces, their normals pointing in, came out dark. With shadows on, bpp's shader turns every normal toward the eye, which hid it. `parts/gearsv50.scad` now lists them clockwise. Checked: all 80 pieces of a gear had negative volume to Manifold (manifold3d) and now have positive; OpenSCAD 2021 makes the same spider and side gears as before (932 and 1256 triangles, the same volumes, watertight); the tour passes, on its own and in the hall.

Meshing

The four 16-tooth mitre gears mesh with each spider turned half a tooth (11.25 degrees) from the side gears, as the geometry says: each side gear has a tooth pointing at each spider, so the spider needs a gap there.

Checked outside Bullet by intersecting the printed meshes (trimesh, manifold3d): the overlap volume is zero between 11.0 and 11.5 degrees, and rises steadily on either side, to 488 mm³ at 0 degrees.

Mesh-to-mesh contact (GImpact) doesn't work for these teeth

With each gear's printed mesh as its collision shape (Bullet's GImpact), the gears never behaved: - **Holding fails:** an input held by its motor was dragged round with the other. - **No clean reading:** the carrier's turn was 10 to 60% off (A + B) / 2. - **Spiders loose:** the spiders spun about axes other than their pins. - **Nothing helped:** changing the margin (0.05 mm), the solver iterations (10 to 50), the motor strength (5 to 500), the friction (0.05 to 0.3), or driving the inputs kinematically. - **Phase search:** looking for the spiders' phase by the kick their teeth gave at rest found only noise: near-zero at isolated angles, large at their neighbours.

Two set-up mistakes were found on the way, then fixed: - An OpenSCAD mesh's place has to be set after it is added to the world. At first all four gears sat at the origin. - A part's collisions are fixed when it is added. Turning `collides` on afterwards does nothing.

Convex pieces do work

Each gear as a compound of convex hulls: its body inside the root cone (224 points), and its 16 teeth outside it (one 24-point hull, placed 16 times). The tooth's hull is only 2% bigger than the tooth, the

body's 1.6%. Masses and inertias come from the printed parts at 0.8 g/cm³. With motors on the inputs, over 4 s:

A (turns/s)	B (turns/s)	carrier (turns)	(A + B) / 2	error
0.25	0 (held)	+0.4751	+0.4751	0.00%
0.25	0.25	+0.9495	+0.9495	0.00%
0.25	-0.25	-0.0009	-0.0009	(2e-5 turns)
0.5	0.1	+1.1461	+1.1461	0.00%
-0.4	0.3	-0.1926	-0.1926	0.02%

That was at 10 solver iterations, and it ran faster than real time. The exhibit uses 20.

The hand

- **Both at once:** turning both inputs together and then stopping one jams the simulated teeth. The load jumps across the backlash all at once. So the hand turns one input at a time, as a person turns one crank at a time.
- **Holding:** an input held only at zero speed creeps (0.08 units) under the load from the other. The hand steers the held one back to where it was set.
- **Back-driving:** at friction 0.1 the held input is pushed 0.3 units before the hand corrects it. That is realistic: a slippery differential back-drives its other input unless it is held firmly.

The tour (3 + 4, then 12, then 10, then 0)

Setting	Result
friction 0.1	PASS
friction 0.3	PASS
friction 0.5	PASS
hand speed 0.2 turns/s	PASS
hand speed 0.4 turns/s	PASS
hand speed 0.8 turns/s	PASS

The ring agrees with A + B to 0.01 units every time.

Resting

Once the hand has nothing to do and nothing has moved for half a second, the Differential stops stepping until there is something to do again. Stepped at 2400 steps a second, a machine at rest costs as much as one running, and its parts never quite stop (their sleeping is turned off, as it must be while they work): what was left once it had settled was up to about 0.05 cm/s and 0.03 rad/s (the gears in their backlash). "Not moving" is set just above that: 0.1 cm/s and 0.05 rad/s. Resting can't change a result: woken, it carries on from exactly the state it stopped in, speeds and all. Its sweep (friction 0.1, 0.3, 0.5; hand 0.2 and 0.8 turns a second) was run again with resting on: 6/6. Over a 70 s run, 79 s of computing became 16 s.